

# ENTERPRISE RESOURCE PLANNING (ERP) SYSTEM

## Functional Requirements Document (FRD) & Technical Requirements Document (TRD)

### Version

1.0

### Prepared For

Enterprise ERP Implementation

### Technology Platform

- Frontend: Next.js
- Backend: Node.js / Next.js API Services
- Database: Microsoft SQL Server
- Mobile: React Native (Android & iOS)
- Analytics: Power BI
- Hosting: Azure / AWS / On-Premise

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## 1. EXECUTIVE SUMMARY

The ERP system shall provide a fully integrated enterprise platform for managing financial operations, fuel operations, inventory, procurement, retail operations, human resources, payroll, maintenance, visitor management, executive analytics, and mobile workforce operations.

The solution shall support:

- Multi-company
- Multi-branch
- Multi-currency
- Multi-bank
- Role-based security
- Workflow approvals
- Audit trails

- Mobile operations
  - Offline synchronization
  - Executive reporting
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## **2. BUSINESS OBJECTIVES**

The ERP shall:

- Centralize enterprise operations
  - Improve financial visibility
  - Enhance operational efficiency
  - Improve inventory accuracy
  - Reduce fraud through workflow controls
  - Enable branch-level profitability analysis
  - Improve maintenance planning
  - Streamline HR and payroll processes
  - Improve visitor and security management
  - Provide executive dashboards and forecasting
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## **3. USER ROLES**

### **System Administrator**

Full system access.

### **Finance Manager**

Finance and accounting functions.

### **Accountant**

Financial transactions and reporting.

### **Procurement Officer**

Purchasing operations.

**Inventory Officer**

Stock management.

**Warehouse Manager**

Warehouse operations.

**Fuel Operations Manager**

Fuel inventory and reconciliation.

**Station Manager**

Retail and station operations.

**HR Manager**

Human resources and payroll.

**Maintenance Manager**

Maintenance planning and execution.

**Technician**

Maintenance execution.

**Receptionist**

Visitor registration.

**Security Officer**

Visitor access control.

**Executive Management**

Dashboards and analytics.

**Employee**

Self-service portal.

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## **4. FUNCTIONAL MODULES**

### **MODULE 1: FINANCIAL CONTROLS**

#### **Features**

##### **Approval Matrix**

- Multi-level approvals
- Approval limits
- Escalations
- Delegations

##### **Segregation of Duties**

- Maker-checker controls
- Role restrictions
- Audit validation

##### **Accrual Management**

- Automatic accrual journals
- Recurring accrual schedules

##### **Intercompany Accounting**

- Intercompany transactions
- Due-to / Due-from accounts
- Elimination entries

##### **Consolidation**

- Branch consolidation
- Company consolidation
- Group reporting

##### **Foreign Currency**

- Exchange rate management
- Currency revaluation

- Forex gain/loss postings

## **Banking**

- Multi-bank support
  - Electronic payments
  - Reconciliations
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## **GENERAL LEDGER**

### **Features**

- Chart of Accounts
- Journal Entries
- Recurring Journals
- Adjusting Entries
- Reversing Entries
- Fiscal Calendar
- Trial Balance
- Period Closing
- Year-End Closing

### **Reports**

- Trial Balance
  - General Ledger
  - Profit & Loss
  - Balance Sheet
  - Cash Flow
  - Financial Ratios
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## **ACCOUNTS RECEIVABLE**

### **Features**

- Customer Master
- Invoicing
- Receipts
- Credit Notes
- Debit Notes
- Statements
- Collections

### **Reports**

- Customer Aging
  - Outstanding Invoices
  - Collection Analysis
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## **ACCOUNTS PAYABLE**

### **Features**

- Supplier Management
- Invoice Processing
- Payments
- Credit Notes
- Statements

### **Reports**

- Supplier Aging
  - Due Payments
  - Vendor Analysis
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## **FIXED ASSETS**

### **Features**

- Asset Register
- Asset Categories
- Depreciation
- Transfers
- Disposal
- Maintenance Tracking

### **Reports**

- Asset Register
  - Depreciation Schedule
  - Asset Movement
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## **BUDGETING**

### **Features**

- Annual Budgets
- Department Budgets
- Revisions
- Approvals

### **Reports**

- Budget vs Actual
  - Variance Analysis
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## **MODULE 2: RETAIL OPERATIONS**

### **Loyalty Program**

- Customer registration
- Point accumulation
- Redemption

### **Gift Cards**

- Card issuance
- Card redemption
- Balance inquiry

### **Coupons**

- Coupon creation
- Coupon validation
- Redemption tracking

### **Promotions**

- Product promotions
- Branch promotions
- Campaign management

### **Dynamic Pricing**

- Rule-based pricing
  - Time-based pricing
  - Customer-specific pricing
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## **MODULE 3: HR & PAYROLL**

### **Employee Management**

- Employee master
- Organizational structure
- Employment history

### **Attendance**

- Biometric integration
- GPS attendance
- Shift management

### **Payroll**

- Salary structures
- Earnings
- Deductions
- Overtime
- Loans
- Payroll approvals

### **Leave Management**

- Leave applications
- Leave approvals
- Leave balances

### **Employee Self-Service**

- Leave requests
  - Payslips
  - Expense claims
  - Personal profile updates
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## **MODULE 4: MAINTENANCE MANAGEMENT**

### **Features**

- Preventive maintenance
- Predictive maintenance
- Corrective maintenance
- Work orders
- Technician assignments
- Service contracts
- Warranty tracking

### **Reports**

- Downtime analysis
  - Maintenance costs
  - Technician productivity
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## **MODULE 5: FUEL OPERATIONS**

### **Tank Management**

- Tank registration
- Tank capacity tracking
- Tank dipping

### **Pump Management**

- Pump registration
- Nozzle management
- Meter readings

### **Fuel Deliveries**

- Delivery recording

- Variance management

### **Reconciliation**

- Daily fuel reconciliation
- Loss/gain analysis
- Margin calculations

### **Reports**

- Tank reconciliation
  - Pump profitability
  - Product profitability
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## **MODULE 6: INVENTORY & PROCUREMENT**

### **Inventory**

- Item master
- Batch management
- Serial management
- Multi-warehouse

### **Procurement**

- Requisitions
- RFQs
- Purchase Orders
- Goods Receipts
- Supplier evaluation

### **Warehouse**

- Bin management
- Transfers

- Cycle counts

### **Valuation**

- FIFO
- Weighted Average

### **Barcode & QR**

- Barcode generation
  - QR scanning
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## **MODULE 7: EXECUTIVE ANALYTICS**

### **KPI Dashboards**

- Revenue
- Gross Profit
- Net Profit
- Inventory Turnover
- Payroll Cost

### **Profitability**

- By Branch
- By Product
- By Station
- By Pump

### **Analytics**

- Heat Maps
  - Trend Analysis
  - Forecasting
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## **MODULE 8: MOBILE APPLICATION**

### **Platforms**

- Android
- iOS

### **Features**

- Offline Mode
  - Synchronization
  - GPS Tracking
  - Camera Authentication
  - Push Notifications
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## **MODULE 9: VISITOR MANAGEMENT**

### **Visitor Registration**

- Walk-in visitors
- Pre-registered visitors
- Visitor categories

### **Visitor Check-In / Check-Out (OPTIONAL)**

- Identity verification
- Visitor photographs
- Visitor badges
- QR code passes

### **Contractor Management (OPTIONAL)**

- Contractor registration
- Insurance tracking
- Safety certification tracking

### **Vehicle Management (OPTIONAL)**

- Visitor vehicle registration
- Delivery vehicle tracking

### **Security Screening (OPTIONAL)**

- Asset declarations
- Restricted area tracking

### **Notifications (OPTIONAL)**

- Visitor arrival alerts
- Approval requests
- Appointment reminders

### **Reports**

- Visitor history
  - Visitors on site
  - Contractor visits
  - Vehicle movement reports
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## **MODULE 10: WORKFLOW & APPROVAL ENGINE**

### **Features**

- Workflow Designer
- Approval Levels
- Escalations
- Delegations
- Notifications
- Audit Logs

Applicable to:

- Purchase Orders
  - Payments
  - Payroll
  - Journal Entries
  - Asset Disposal
  - Leave Requests
  - Budget Approvals
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## **MODULE 11: SECURITY & ADMINISTRATION**

### **Features**

- User Management
  - Role Management
  - Permission Management
  - MFA Authentication
  - Audit Trails
  - Password Policies
  - Session Management
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## **REPORTING REQUIREMENTS**

The ERP shall provide:

- Operational Reports
- Financial Reports
- Compliance Reports
- Executive Dashboards
- Scheduled Reports
- Export to PDF/Excel

## DIAGRAM

